Practical-2: Intermediate Programs Using User-Defined Function

1) Return the maximum of three floating-point numbers. (A)

Program :-

```
#include <stdio.h>
float maxOfThree(float ,float ,float);
void main(){
	float n1,n2,n3;
	printf("Enter the first Number : ");
	scanf("%f",&n1);
	printf("Enter the second Number : ");
	scanf("%f",&n2);
	printf("Enter the third Number : ");
	scanf("%f",&n3);
	float max = maxOfThree(n1,n2,n3);
	printf("Max = %.2f",max); //up to two decimal point round of
}
float maxOfThree(float n1,float n2,float n3){
	if(n1 > n2){
		if(n1 > n3){
			return n1;
		}
		else{
			return n3;
		}
	}
	else{
		if(n2 > n3){
			return n2;
		}
		else{
			return n3;
		}
	}
}
```

Output :-

```
Enter the first Number : 23.45
Enter the second Number : 35.67
Enter the third Number : 35.45
Max = 35.67
```

2) Return the minimum of three floating-point numbers.?(A)

Program :-

```c
#include <stdio.h>
float minOfThree(float ,float ,float);
void main(){
    float n1,n2,n3;
    printf("Enter the first Number : ");
    scanf("%f",&n1);
    printf("Enter the second Number : ");
    scanf("%f",&n2);
    printf("Enter the third Number : ");
    scanf("%f",&n3);
    float max = minOfThree(n1,n2,n3);
    printf("Min = %.2f",max); //up to two decimal point round of
}
float minOfThree(float n1,float n2,float n3){
    if(n1 < n2){
        if(n1 < n3){
            return n1;
        }
        else{
            return n3;
        }
    }
    else{
        if(n2 < n3){
            return n2;
        }
        else{
            return n3;
        }
    }
}
```

Output :-

```
Enter the first Number : 23.45
Enter the second Number : 35.67
Enter the third Number : 35.45
Min = 23.45
```

3) Swap two numbers using call by value and call by reference.?(A)

Program :-

```
#include <stdio.h>
void swapByCallByValue(int, int);
void swapByCallByReference(int*, int*);
void main(){
	int n1,n2;
	printf("Enter the n1 : ");
	scanf("%d",&n1);
	printf("Enter the n2 : ");
	scanf("%d",&n2);
	//call by value
	//can not change in Actual Parameter
	swapByCallByValue(n1,n2);
	//call by reference
	//also change in Actual Parameter
	swapByCallByReference(&n1,&n2);
	printf("call by reference \n");
	printf("n1 = %d and n2 = %d \n",n1,n2);
}

void swapByCallByValue(int n1, int n2){
	//formal Parameter
	int temp = n1;
	n1 = n2;
	n2 = temp;
	printf("call by value \n");
	printf("n1 = %d and n2 = %d \n",n1,n2);
}

void swapByCallByReference(int* p1, int* p2){
	int temp = *p1;
	*p1 = *p2;
	*p2 = temp;
}
```

Output :-

```
Enter the n1 : 12
Enter the n2 : 25
call by value
n1 = 25 and n2 = 12
call by reference
n1 = 25 and n2 = 12
```

4) Find all prime numbers between given interval using functions.?(B)

Program :-

```
#include<stdio.h>
void findPrime(int ,int);
int isPrime(int);
void main(){
        int start,end;
        printf("Enter the start : ");
        scanf("%d",&start);
        printf("Enter the end : ");
        scanf("%d",&end);
        findPrime(start,end);
}
void findPrime(int start,int end){
        int i = 0;
        for(i=start;i<=end;i++){
                if(isPrime(i)==1){
                        printf("%d \n",i);
                }
                else{
                        continue;
                }
        }
}
int isPrime(int n){
        int count = 0;
        int i = 0;
        for(i=1;i<=n;i++){
                if(n%i==0){
                        count++;
                }
        }
        if(count == 2){
                return 1;
        }
        else{
                return 0;
        }
}
```

Output :-

```
Enter the start : 1
Enter the end : 20
2
3
5
7
11
13
17
19
```

5) Create a function that converts amount into words.
(i.e. 9241: Nine Thousand Two Hundred Forty One)?(C)

Program :-

```
#include <stdio.h>
#include <string.h>

void number_to_words(int);

void main() {
   int n = 0;
   printf("Enter the Four Digit Number : ");
   scanf("%d",&n);
   number_to_words(n);
}

void number_to_words(int n) {
    char result[1024] = "";
   // Arrays for number words
   char *ones[] = { "", "One", "Two", "Three", "Four", "Five", "Six", "Seven", "Eight", "Nine", "Ten"};
   char *teens[] = {"Ten","Eleven", "Twelve", "Thirteen", "Fourteen", "Fifteen", "Sixteen",
"Seventeen", "Eighteen", "Nineteen" };
   char *tens[] = { "", "", "Twenty ", "Thirty ", "Forty ", "Fifty ", "Sixty ", "Seventy ", "Eighty ",
"Ninety " };

        if (n == 0) {
                printf("Zero");
                return;
        }
        if(n >= 1000){
                strcat(result,ones[n/1000]);
                strcat(result," Thousand ");
                n = n % 1000;
        }
        if(n >= 100){
                strcat(result,ones[n/100]);
                strcat(result," Hundred ");
                n = n % 100;
        }
        if(n >= 20){
                strcat(result,tens[n/10]);
                n = n % 10;
                strcat(result,ones[n]);
                n = 0;
        }
        if(n >= 10){
                strcat(result,teens[n%10]);
                n = 0;
        }
```

```
        if(n > 0){
                strcat(result,ones[n]);
                n = 0;
        }
        puts(result);
}
```

Output :-

```
Enter the Four Digit Number : 9241
Nine Thousands Two Hundreds Forty One
```